

INTEGRAL EQUIVARIANT K-THEORY OF THE INERT LOCAL MODEL: THE RAY-CLASS ENTANGLEMENT AND EXACT TORSION

RUQING CHEN

ABSTRACT. We carry out the first integral case of Paper XXXII's Problem (i): the equivariant K-theory of the inert local model at $K = \mathbb{Q}(\sqrt{-5})$, $H = \mathbb{Q}(i, \sqrt{5})$, $\mathfrak{p}_3 = (3, 1 + \sqrt{-5})$ inert in H/K , $C = \text{Gal}(H/K) \cong \mathbb{Z}/2$. The ring $\mathcal{O}_H = \mathbb{Z}[i] \otimes \mathbb{Z}[\varphi]$ is a *free* $\mathbb{Z}[C]$ -module of rank two (a Tate cohomology computation), so the entire exterior tower carries explicit integral C -structure: $\Lambda^0 = \mathbb{Z}_+$, $\Lambda^1 \cong \mathbb{Z}[C]^2$, $\Lambda^2 \cong \mathbb{Z}_-^2 \oplus \mathbb{Z}[C]^2$, $\Lambda^3 \cong \mathbb{Z}[C]^2$, $\Lambda^4 = \mathbb{Z}_+$. A finite search produces the explicit generator $\Pi = -3 - i + 2\varphi + i\varphi$ of a prime above 3, with $N_{H/\mathbb{Q}}\Pi = +9$ and $\text{Tr}_{H/\mathbb{Q}}\Pi = -8$, and Smith normal forms of $9 \cdot 1 - \Lambda^r\Pi$ evaluate the intermediate layers exactly (prime-to-3 regularization): $\mathbb{Z}/5 \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/61$ at $r = 1$; $\mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/7 \oplus \mathbb{Z}/11$ at $r = 2$, containing the sign-twisted $\mathbb{Z}_-^2$ -block; $\mathbb{Z}/29$ at $r = 3$. Assembling with the top and bottom layers gives

$$K_0 \supseteq \mathbb{Z}^2 \oplus (\mathbb{Z}/8)^2 \oplus [\mathbb{Z}/2^{\oplus 2} \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/7 \oplus \mathbb{Z}/11], \quad K_1 \supseteq \mathbb{Z}^2 \oplus [\mathbb{Z}/5^{\oplus 2} \oplus \mathbb{Z}/61] \oplus \mathbb{Z}/29,$$

where $(\mathbb{Z}/8)^2 = \mathbb{Z}/(q^h - 1) \otimes R(\mathbb{Z}/2)$: the unit class is split by the spectral projections $\frac{1 \pm U_\sigma}{2}$ into a pair of classes of exact order 8. The entanglement law this computation exhibits: the class group doubles or sign-twists exactly the *non-induced* strata, while free $\mathbb{Z}[C]$ -blocks count once by Shapiro — the class group does not change the orders of the middle layers, only their representation labels. Two honest gaps close the paper: the u_σ -twisted bookkeeping of those labels, and the extension data in degree four. A structural by-product is recorded: the σ -fixed locus of the inert model is the bottom h -step system of Paper XXXI — null in measure, nonempty in topology — so Paper XXXII's Problem (ii) is a direct summand of Problem (i), not its sequel.

1. THE EXPLICIT GENERATOR AND THE EQUIVARIANT EXTERIOR TOWER

The discriminants -4 and 5 are coprime, so $\mathcal{O}_H = \mathbb{Z}[i] \otimes \mathbb{Z}[\varphi]$, $\varphi = \frac{1+\sqrt{5}}{2}$, with σ (the generator of C , fixing $\sqrt{-5} = i\sqrt{5}$) acting by $i \mapsto -i$, $\varphi \mapsto 1 - \varphi$.

Lemma 1.1. $\mathbb{Z}[\varphi] \cong \mathbb{Z}[C]$ as a C -module ($\hat{H}^0 = \hat{H}^1 = 0$: the norm has image $\mathbb{Z}$ and $\ker N = \text{im}(1 - \sigma) = \mathbb{Z}\sqrt{5}$), and $\mathbb{Z}[i] \cong \mathbb{Z}_+ \oplus \mathbb{Z}_-$; hence $\mathcal{O}_H \cong \mathbb{Z}[C]^2$ is free, and the integral exterior tower is

$$\Lambda^0 = \mathbb{Z}_+, \quad \Lambda^1 \cong \mathbb{Z}[C]^2, \quad \Lambda^2 \cong \mathbb{Z}_-^2 \oplus \mathbb{Z}[C]^2, \quad \Lambda^3 \cong \mathbb{Z}[C]^2, \quad \Lambda^4 = \mathbb{Z}_+,$$

the $\mathbb{Z}_-$'s in Λ^2 coming from $\Lambda_{\mathbb{Z}}^2(\mathbb{Z}[C])$ ($\sigma(1 \wedge \sigma) = -(1 \wedge \sigma)$), and $\det \sigma = +1$ on Λ^4 .

Proposition 1.2 (The generator). $\Pi = -3 - i + 2\varphi + i\varphi \in \mathcal{O}_H$ satisfies $N_{H/\mathbb{Q}}\Pi = +9$ and $\text{Tr}_{H/\mathbb{Q}}\Pi = -8$; its multiplication matrix A on the basis $(1, i, \varphi, i\varphi)$ has $\det A = +9$, so $(\Pi) =: \mathfrak{P}_1$ is a prime of norm 9 above 3, and the inert model of Paper XXXII at $\mathfrak{P}_1$ is defined with scaling Π .

Date: 6 September 2026.

Paper XXXVIII of the series on localized Bost–Connes systems. The framework is that of [1], cited by DOI; Papers XXX, XXXI and XXXII of the series are referred to by numeral, and every fact used is restated in place. Search and Smith-form scripts in `code/xxxviii-inert/` of the repository.

2. EXACT EVALUATION OF THE INTERMEDIATE LAYERS

The tower transfers invert only the prime 3 (they are built from the Π -adjugates, of determinant a power of $N\Pi = 9$), so the regularized cokernel of $1 - \alpha_*$, $\alpha_* = 9^{-1}\Lambda^r\Pi$, is the prime-to-3 part of the integral cokernel of $9 \cdot 1 - \Lambda^r A$, computed by Smith normal form:

Theorem 2.1. *With T_r the regularized cokernel at layer r :*

$$\begin{aligned} r = 1 : \quad & \det(9 - \Lambda^1 A) = 13725 = 3^2 \cdot 5^2 \cdot 61, \quad T_1 \cong \mathbb{Z}/5 \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/61; \\ r = 2 : \quad & \det(9 - \Lambda^2 A) = -1122660, \quad T_2 \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/7 \oplus \mathbb{Z}/11; \\ r = 3 : \quad & \det(9 - \Lambda^3 A) = 21141 = 3^6 \cdot 29, \quad T_3 \cong \mathbb{Z}/29, \end{aligned}$$

T_2 containing the contribution of the sign-twisted $\mathbb{Z}_-^2$ -block of Lemma 1.1. At the ends, $T_0 = \mathbb{Z}/(9 - 1) = \mathbb{Z}/8$ generated by the unit class, and $M_4 = \mathbb{Z}$ with $\alpha_* = N\Pi/9 = +1$.

3. THE K-THEORY OF THE CROSSED PRODUCT AND THE ENTANGLEMENT LAW

Theorem 3.1. *For the inert local model at $\mathfrak{P}_1$ with its $\mathbb{Z}/2$ -action,*

$$K_0(\partial\mathcal{A}_{\mathfrak{P}_1}^{\text{aff,loc}} \rtimes \mathbb{Z}/2) \supseteq \mathbb{Z}^2 \oplus (\mathbb{Z}/8)^2 \oplus [\mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/7 \oplus \mathbb{Z}/11],$$

$$K_1(\partial\mathcal{A}_{\mathfrak{P}_1}^{\text{aff,loc}} \rtimes \mathbb{Z}/2) \supseteq \mathbb{Z}^2 \oplus [\mathbb{Z}/5 \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/61] \oplus \mathbb{Z}/29.$$

Here $(\mathbb{Z}/8)^2 = T_0 \otimes R(\mathbb{Z}/2)$: the unit class is split by the spectral projections $\frac{1 \pm U_\sigma}{2}$ of the implementing unitary into two classes of exact order 8, one per character of the class group; and the two $\mathbb{Z}$'s are the $R(\mathbb{Z}/2)$ -doubling of the trivially-acted top layer.

Remark 3.2 (The entanglement law). The class group entangles with the q -torsion precisely on the *non-induced* strata: layers with trivial or sign action ($r = 0$, $r = 4$, and the $\mathbb{Z}_-^2$ -block of Λ^2) are doubled or sign-twisted by $R(\mathbb{Z}/2)$, while the free $\mathbb{Z}[C]$ -blocks contribute once, by Shapiro's lemma. The class group does not change the orders of the middle layers — those are the Smith invariants of Theorem 2.1 — it changes their representation labels.

Remark 3.3 (Problem (ii) is a summand of Problem (i)). The σ -fixed locus of $\mathcal{O}_{\mathfrak{P}_1}$ is $\mathcal{O}_{\mathfrak{P}_3}$: null for the Gibbs measure, nonempty in topology — and K-theory sees topology. The fixed-point contribution to the equivariant theory is the bottom h -step system of Paper XXXI re-entering as the fixed subgroupoid: the fixed-point-versus-crossed-product question of Paper XXXII is a direct summand of the present computation, not its sequel.

4. ASSESSMENT

Two gaps remain and are recorded as such. (a) *Character bookkeeping*: the scaling is equivariant only up to the descent cocycle $u_\sigma = \sigma\Pi/\Pi$ of Paper XXXII, so the middle layers are twisted $R(\mathbb{Z}/2)$ -modules; their orders are fixed by Theorem 2.1, but the assignment of each cyclic summand to the trivial or sign character awaits the twisted bookkeeping. (b) *Extension data*: in degree four the Pimsner–Voiculescu assembly determines K_* up to group extensions, as in Paper XXX. Neither gap touches the displayed subgroups, the split unit class, or the entanglement law.

REFERENCES

- [1] R. Chen, *KMS state uniqueness and phase transitions in localized Bost–Connes systems*, preprint, 2026, <https://doi.org/10.5281/zenodo.22152101>.
- [2] C. W. Curtis, I. Reiner, *Methods of Representation Theory I*, Wiley, 1981.

GUT GEOSERVICE INC., MONTREAL, CANADA
Email address: ruqing@hotmail.com